Lecture 28.05: Entropy vs Gini Impurity — Mathematical Comparison & Invariants

📌 Executive Summary

Rigorous mathematical comparison of Entropy vs Gini, Taylor series connection, split selection concordance rates, and computational throughput considerations.

🏛️ Theoretical Formulation & Mathematical Architecture

🔬 The Taylor Series Mathematical Equivalence

While Shannon Entropy derives from information theory and Gini Impurity derives from variance of multinomial indicators, they are mathematically connected through the **first-order Taylor series expansion** of the natural logarithm:

$$\ln(x) = -\sum_{n=1}^{\infty} \frac{(1-x)^n}{n} \approx -(1-x) = x - 1 \quad \text{for } x \approx 1$$

Expressing Entropy in natural log base:

$$H_e(S) = -\sum_{k=1}^{K} p_k \ln(p_k)$$

Substituting $\ln(p_k) \approx p_k - 1$:

$$H_e(S) \approx -\sum_{k=1}^{K} p_k(p_k - 1) = \sum_{k=1}^{K}(p_k - p_k^2) = 1 - \sum_{k=1}^{K} p_k^2 = G(S)$$

Thus, **Gini Impurity is the first-order Taylor polynomial approximation of Shannon Entropy**.

⚖️ Side-by-Side Architectural Summary

Metric Feature	Shannon Entropy $H(S)$	Gini Impurity $G(S)$
Formula	$-\sum p_k \log_2 p_k$	$1 - \sum p_k^2$
Binary Range	$[0.0, 1.0]$ bits	$[0.0, 0.5]$
Peak (Binary)	1.0 at $p = 0.5$	0.5 at $p = 0.5$
Computation Cost	Higher (calls `log2`, SIMD transcendentals)	Minimal (pure algebraic multiply-accumulate)
Default In	ID3, C4.5, C5.0	CART, Scikit-Learn `DecisionTreeClassifier`
Split Selection	Slightly favors balanced splits	Slightly favors isolating dominant pure classes

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💻 Vectorized Python Implementation

```
import numpy as np

p = np.linspace(0.01, 0.99, 100)
entropy = -p * np.log2(p) - (1-p) * np.log2(1-p)
gini = 2 * p * (1-p)
scaled_entropy = 0.5 * entropy

# Measure absolute maximum deviation between scaled entropy and Gini
max_diff = np.max(np.abs(scaled_entropy - gini))
print(f"Max absolute divergence between 0.5*Entropy and Gini: {max_diff:.4f}")
assert max_diff < 0.08, "Scaled entropy and Gini must be tightly aligned!"
```

🔬 Key Engineering Takeaways & Invariants

1. **Axis-Aligned Partitioning**: Decision trees recursively split feature space along orthogonal boundaries without requiring feature scaling.
2. **Impurity Optimization**: Split selection deterministically maximizes impurity reduction (Information Gain for classification, Variance Reduction for regression).

3. **Generalization Safeguards**: Regularization via pre-pruning (`max_depth` , `min_samples_leaf`) and post-pruning (`ccp_alpha`) prevents noisy leaf memorization.